

Seminars 3 and 4: Spectrum Representation

Questions and Answers

1. From a sum of sinusoids to a two-sided spectrum

Consider

$$x(t) = 4 + 6\cos(2\pi 100t + \pi/3) + 3\sin(2\pi 250t).$$

(a) Rewrite every sinusoidal term as a sum of complex exponentials.

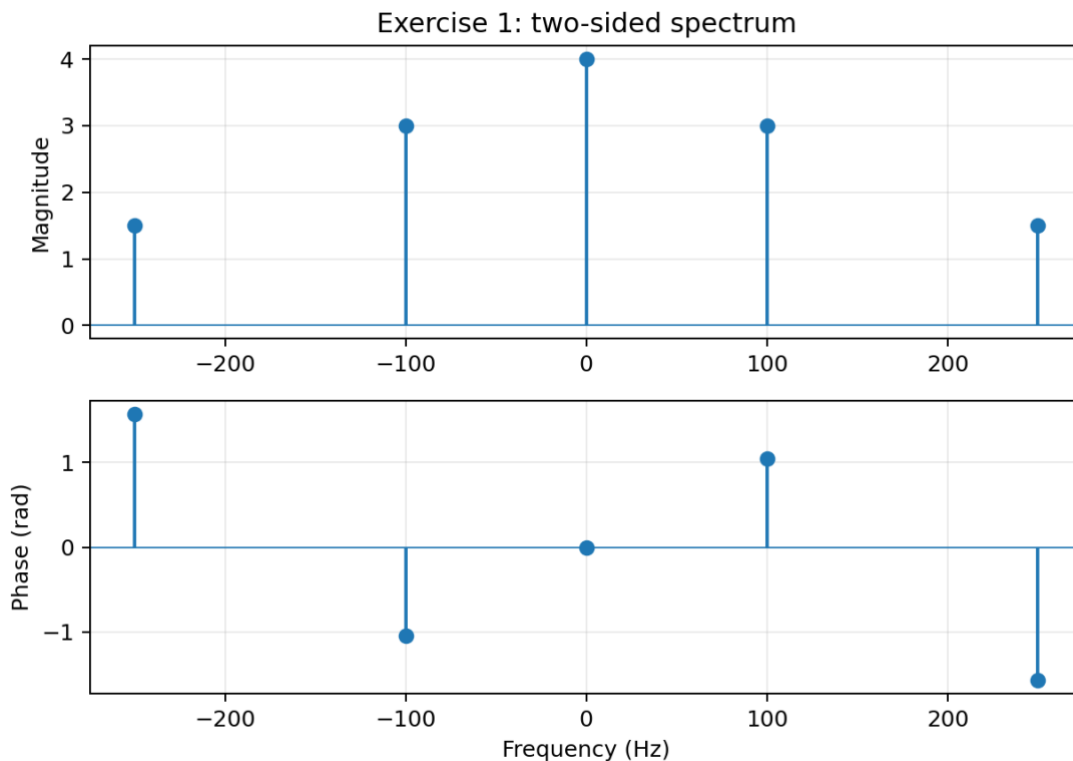
Answer. $6\cos(2\pi 100t + \pi/3) = 3e^{j\pi/3}e^{j2\pi 100t} + 3e^{-j\pi/3}e^{-j2\pi 100t}$ and $3\sin(2\pi 250t) = 1.5e^{-j\pi/2}e^{j2\pi 250t} + 1.5e^{j\pi/2}e^{-j2\pi 250t}$.

(b) List all frequency-complex-amplitude pairs (f_k, X_k) , including DC and negative frequencies.

Answer. $(-250, 1.5e^{j\pi/2})$, $(-100, 3e^{-j\pi/3})$, $(0, 4)$, $(100, 3e^{j\pi/3})$, $(250, 1.5e^{-j\pi/2})$.

(c) Draw separate two-sided plots of magnitude and phase.

Answer. The labelled magnitude and phase plots are shown below.



Magnitude and phase spectrum

(d) Verify the conjugate-symmetry relation required for a real signal.

Answer. $X(-f) = X^*(f)$ for both frequency pairs; the DC coefficient is real.

(e) Determine whether $x(t)$ is periodic. If it is, find its fundamental frequency and period.

Answer. Yes. $f_0 = \gcd(100, 250) = 50$ Hz and $T_0 = 1/f_0 = 0.02$ s.

2. From a spectrum to a time-domain signal

The spectrum has a DC line of 11, lines of magnitude 7 at ± 50 Hz with phases $\mp\pi/3$, and lines of magnitude 4 at ± 175 Hz with phases $\mp\pi/2$.

(a) Write $x(t)$ as a sum of real cosines.

Answer. $x(t) = 11 + 14\cos(2\pi 50t - \pi/3) + 8\cos(2\pi 175t - \pi/2)$.

(b) Identify the DC value.

Answer. The DC value is 11.

(c) Determine whether $x(t)$ is real-valued and explain how the spectrum confirms this.

Answer. Yes; the negative-frequency coefficient is the complex conjugate of the positive-frequency coefficient for each pair.

(d) Determine the fundamental frequency and fundamental period.

Answer. $f_0 = \gcd(50, 175) = 25$ Hz and $T_0 = 0.04$ s.

(e) Explain why the negative-frequency lines are necessary.

Answer. They form conjugate pairs with the positive-frequency lines so that the sum is real.

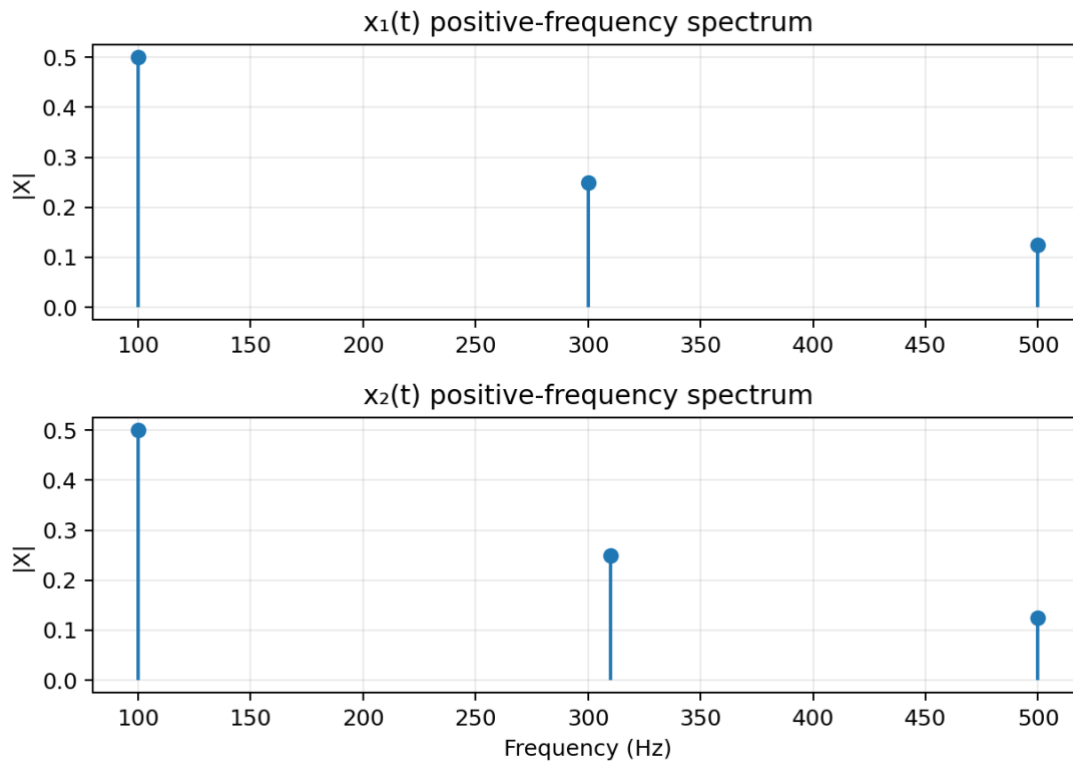
3. Harmonic and nonharmonic sums

$$x_1(t) = \cos(2\pi 100t) + 0.5\cos(2\pi 300t) + 0.25\cos(2\pi 500t),$$

$$x_2(t) = \cos(2\pi 100t) + 0.5\cos(2\pi 310t) + 0.25\cos(2\pi 500t).$$

(a) Sketch the positive-frequency magnitude spectrum of each signal.

Answer. The spectra are shown below; each positive-frequency complex amplitude is half the cosine amplitude.



Positive-frequency spectra

(b) Determine whether each signal is periodic.

Answer. Both are periodic.

(c) For every periodic signal, determine the fundamental frequency and period.

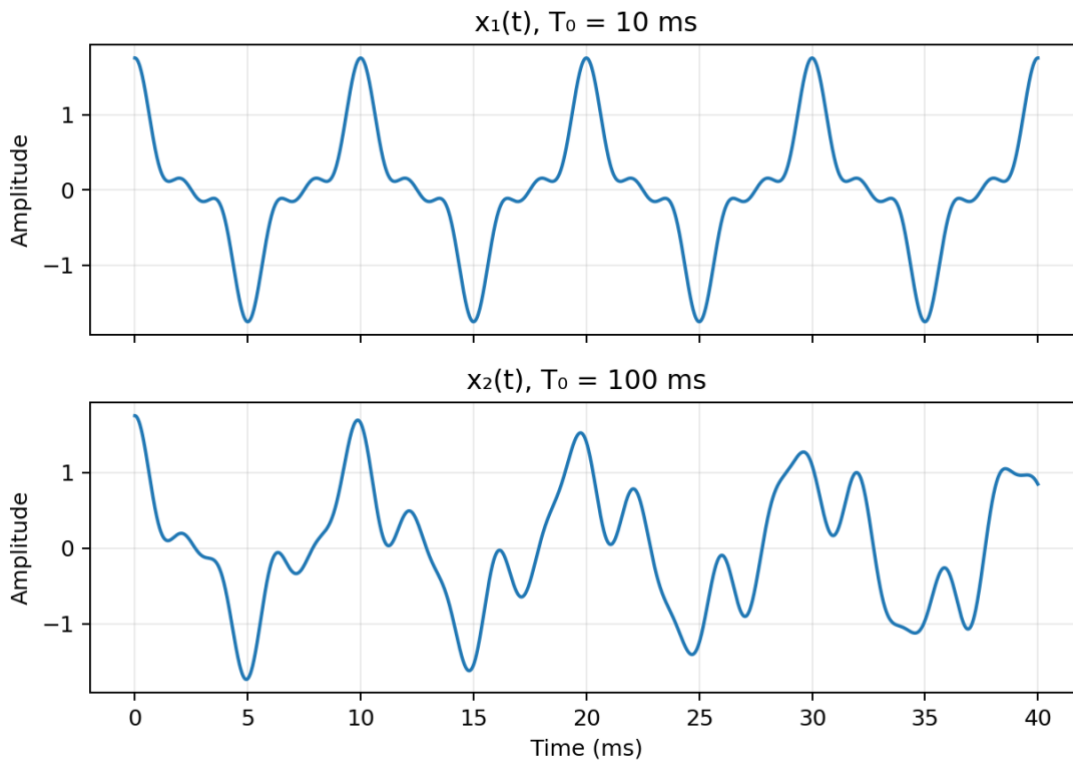
Answer. For x_1 , $f_0 = 100$ Hz and $T_0 = 10$ ms. For x_2 , $f_0 = 10$ Hz and $T_0 = 100$ ms.

(d) Explain the relation between harmonic spacing and periodicity.

Answer. A sum is periodic when all frequencies are integer multiples of a common fundamental frequency.

(e) Plot both signals over 40 ms and compare their repetition patterns.

Answer. The plots are shown below. x_1 repeats every 10 ms; x_2 does not complete its 100-ms period in the displayed interval.



Waveform comparison

4. Beat notes produced by adding close frequencies

$$x(t) = \cos(2\pi 440t) + \cos(2\pi 446t).$$

(a) Express $x(t)$ as a high-frequency cosine multiplied by a slowly varying envelope.

Answer. $x(t) = 2\cos(2\pi 3t)\cos(2\pi 443t)$.

(b) Determine the centre frequency f_c and deviation frequency f_d .

Answer. $f_c = 443 \text{ Hz}$ and $f_d = 3 \text{ Hz}$.

(c) Determine the spectral components.

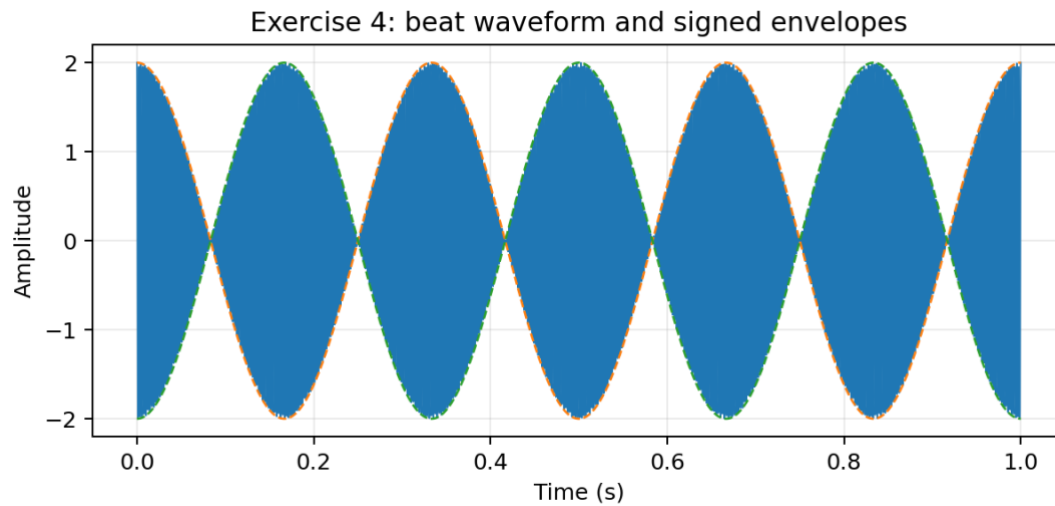
Answer. Lines occur at ± 440 and $\pm 446 \text{ Hz}$, each with magnitude 0.5 and phase 0.

(d) Determine the perceived beat rate.

Answer. The beat rate is 6 beats/s.

(e) Sketch the waveform and both envelope curves over one second.

Answer. The waveform and the signed envelopes are shown below.



Beat waveform

5. Spectrum of a product of sinusoids

$$x(t) = 3\cos(2\pi 50t)\cos(2\pi 400t).$$

(a) Rewrite $x(t)$ as a sum of sinusoids.

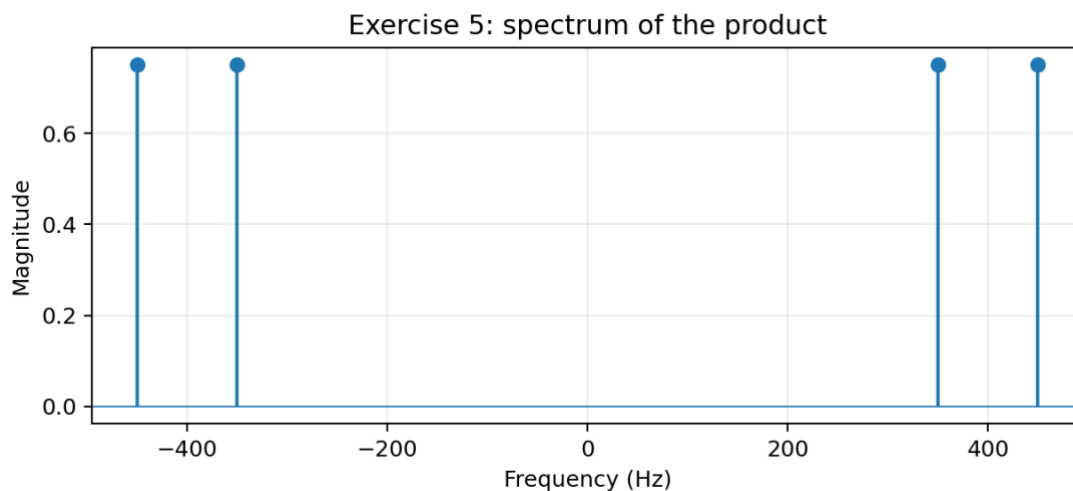
Answer. $x(t) = 1.5\cos(2\pi 350t) + 1.5\cos(2\pi 450t)$.

(b) Determine all spectral-line frequencies and complex amplitudes.

Answer. At ± 350 and ± 450 Hz, each complex amplitude is 0.75.

(c) Sketch the two-sided magnitude spectrum.

Answer. The spectrum is shown below.



Product spectrum

(d) Explain why neither 50 Hz nor 400 Hz appears as a spectral line.

Answer. Multiplication shifts frequency components to the sum and difference frequencies.

(e) Determine periodicity and the fundamental period.

Answer. $f_0 = \gcd(350, 450) = 50$ Hz and $T_0 = 0.02$ s.

6. Amplitude modulation

$$x(t) = [2 + 0.6\cos(2\pi 120t)]\cos(2\pi 1000t).$$

(a) Expand $x(t)$ as a sum of three cosines.

Answer. $x(t) = 2\cos(2\pi 1000t) + 0.3\cos(2\pi 880t) + 0.3\cos(2\pi 1120t)$.

(b) Identify the carrier and sideband frequencies.

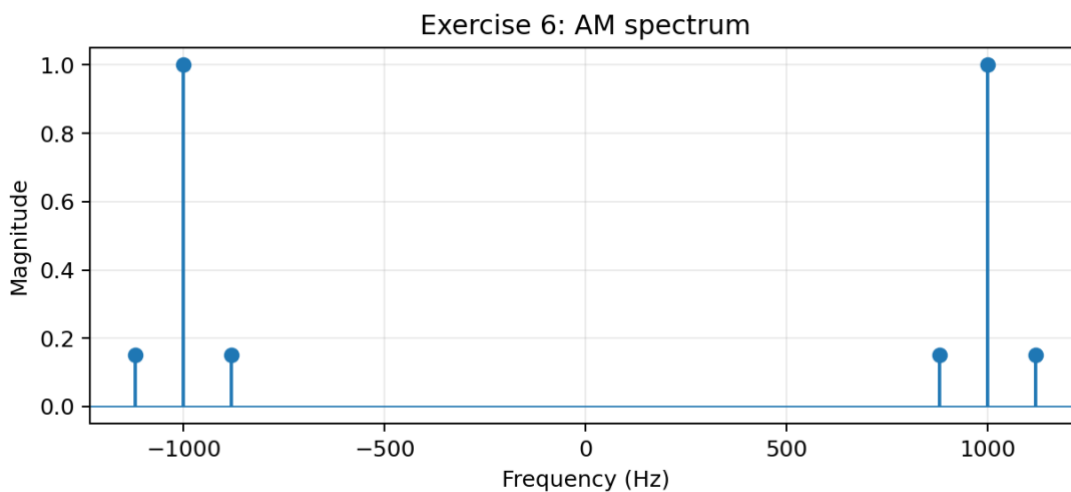
Answer. Carrier: 1000 Hz. Sidebands: 880 and 1120 Hz.

(c) Determine the complex amplitudes at every positive and negative frequency.

Answer. $X(\pm 1000) = 1$ and $X(\pm 880) = X(\pm 1120) = 0.15$, all with zero phase.

(d) Draw the two-sided magnitude spectrum.

Answer. The labelled AM spectrum is shown below.



AM spectrum

(e) State whether the envelope reaches zero and compare with a beat note.

Answer. No. The envelope ranges from 1.4 to 2.6.

7. Finite Fourier-series synthesis

$$x(t) = 10 + 20\cos(2\pi 100t + \pi/4) + 10\sin(2\pi 250t).$$

(a) Find f_0 and T_0 .

Answer. $f_0 = 50$ Hz, $T_0 = 0.02$ s, and $\omega_0 = 100\pi$ rad/s.

(b) Express the signal as $x(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$.

Answer. Use $k = 0, \pm 2, \pm 5$ with the coefficients given in part (c).

(c) Determine the nonzero coefficients and indices.

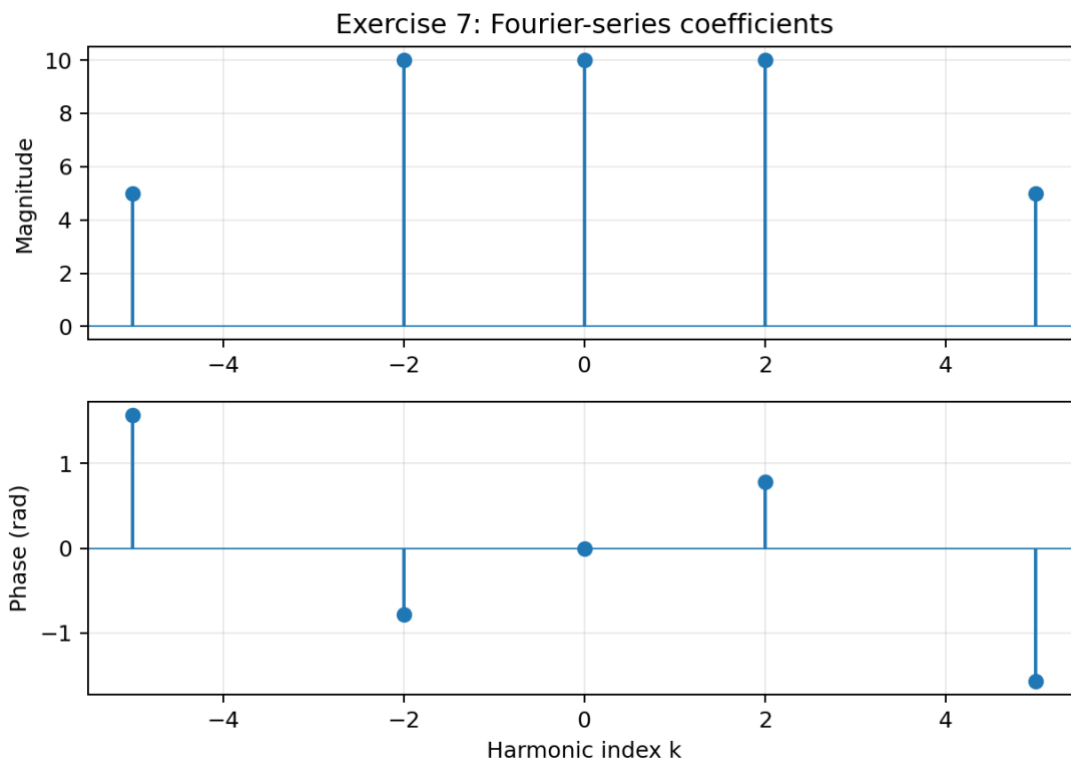
Answer. $a_0 = 10$, $a_2 = 10e^{j\pi/4}$, $a_{-2} = 10e^{-j\pi/4}$, $a_5 = 5e^{-j\pi/2} = -5j$, $a_{-5} = 5e^{j\pi/2} = 5j$.

(d) State a suitable value of N .

Answer. $N = 5$.

(e) Draw the Fourier-series spectrum.

Answer. The magnitude and phase spectra versus harmonic index are shown below.



Fourier-series spectrum

8. Fourier analysis and synthesis equations

A periodic signal has period T_0 and $\omega_0 = 2\pi/T_0$.

(a) Write the synthesis equation.

Answer. $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$.

(b) Write the analysis equation.

Answer. $a_k = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\omega_0 t} dt$.

(c) Show that a_0 is the average value.

Answer. $a_0 = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) dt$.

(d) State conjugate symmetry for a real signal.

Answer. $a_{-k} = a_k^*$.

(e) Explain Fourier analysis versus synthesis.

Answer. Analysis extracts coefficients from a waveform; synthesis reconstructs the waveform from the coefficients.

9. Fourier series of a rectangular pulse train

$$x(t) = \begin{cases} A, & |t| \leq \tau/2, \\ 0, & \tau/2 < |t| \leq T_0/2, \end{cases}$$

extended periodically with period T_0 .

(a) Compute a_0 .

Answer. $a_0 = A\tau/T_0$.

(b) Derive a_k for $k \neq 0$.

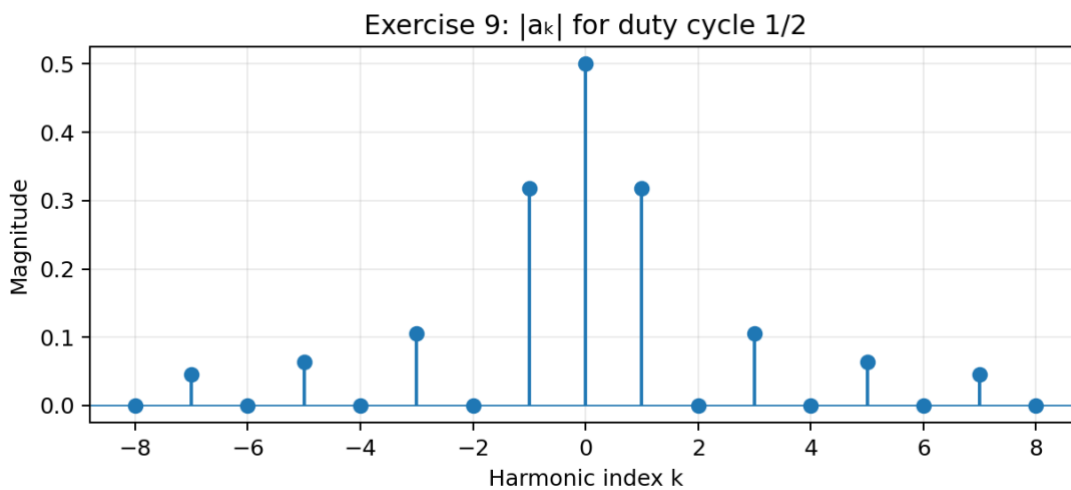
Answer. $a_k = \frac{A}{T_0} \frac{2\sin(k\omega_0\tau/2)}{k\omega_0} = \frac{A\sin(\pi k\tau/T_0)}{\pi k}$.

(c) Express the result using $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.

Answer. $a_k = A \frac{\tau}{T_0} \text{sinc}\left(k \frac{\tau}{T_0}\right)$, valid also at $k = 0$ by continuity.

(d) Sketch $|a_k|$ for duty cycle $1/2$ and $-8 \leq k \leq 8$.

Answer. The coefficient magnitudes are shown below.



Rectangular pulse coefficients

(e) Identify which harmonics vanish and explain why.

Answer. All nonzero even harmonics vanish.

10. Reconstructing a rectangular waveform

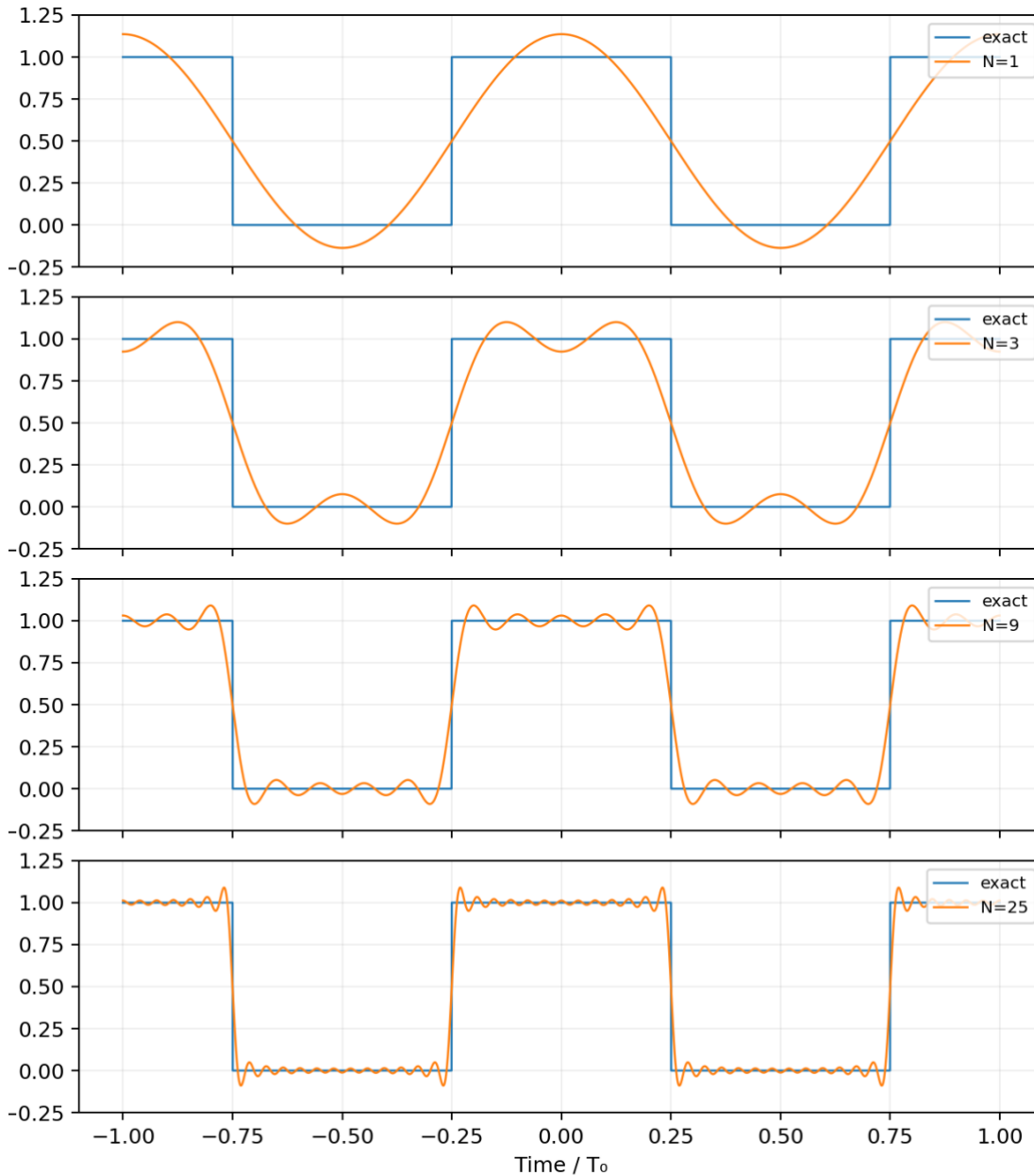
Use the coefficients from Exercise 9 with duty cycle $1/2$.

(a) Construct approximations for $N = 1, 3, 9, 25$.

Answer. $x_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t} = a_0 + 2 \sum_{k=1}^N a_k \cos(k\omega_0 t)$.

(b) Plot the four approximations with the exact waveform.

Answer. The four reconstructions are shown below.



Fourier reconstruction

(c) Describe the change as N increases.

Answer. Edges become sharper and flat regions become more accurate.

(d) Identify where the largest errors occur.

Answer. Near the jump discontinuities.

(e) Relate coefficient decay to sharp discontinuities.

Answer. The $1/|k|$ decay is slow, so many harmonics are required.

11. Spectrum of a nonlinear sinusoidal expression

$$x(t) = \cos^3(\omega_0 t).$$

(a) Express $x(t)$ as a sum of cosines.

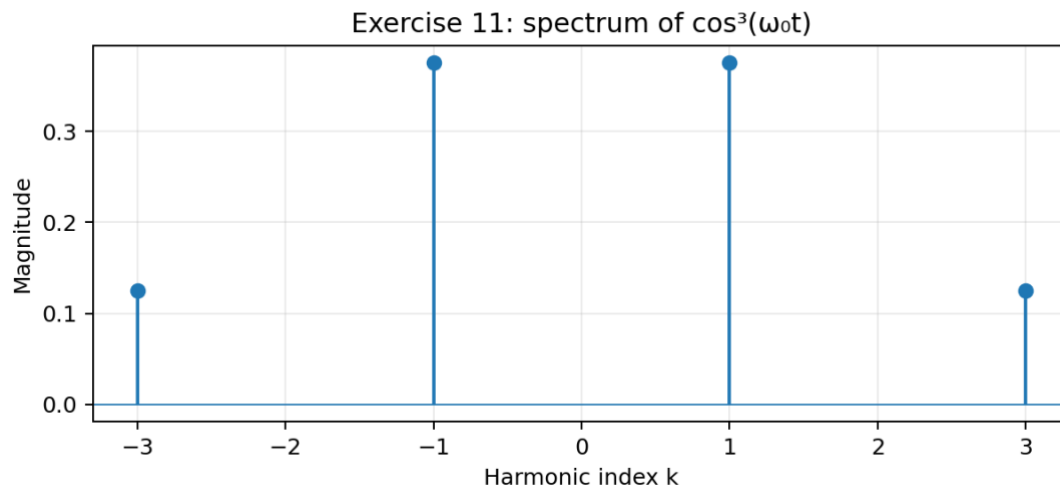
Answer. $\cos^3 \theta = \frac{3}{4} \cos \theta + \frac{1}{4} \cos 3\theta$.

(b) Determine the nonzero Fourier-series coefficients.

Answer. $a_{\pm 1} = 3/8$ and $a_{\pm 3} = 1/8$.

(c) Draw the spectrum versus harmonic index.

Answer. The spectrum is shown below.



Cubed-cosine spectrum

(d) State which harmonics are absent.

Answer. DC and all harmonics except $k = \pm 1, \pm 3$ are absent.

(e) Verify conjugate symmetry.

Answer. All nonzero coefficients are real and satisfy $a_{-k} = a_k = a_k^*$.

12. Operations on periodic signals and Fourier coefficients

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}.$$

(a) $y(t) = Cx(t)$.

Answer. $b_k = C a_k$.

(b) $y(t) = x(t - t_1)$.

Answer. $b_k = a_k e^{-jk\omega_0 t_1}$.

(c) $y(t) = x(-t)$.

Answer. $b_k = a_{-k}$.

(d) $y(t) = dx(t)/dt$.

Answer. $b_k = jk\omega_0 a_k$.

(e) $y(t) = x(t) + x(t - T_0/2)$.

Answer. $b_k = a_k [1 + (-1)^k]$; odd harmonics cancel and even harmonics double.

13. Reading and interpreting a spectrogram

A recording contains, in order: a constant pure tone, a rising chirp, two simultaneous harmonic tones, and a short broadband impulse.

(a) Describe the expected appearance of each event.

Answer. Pure tone: one horizontal line. Chirp: an upward diagonal ridge. Harmonic tones: two stacks of horizontal harmonic lines. Impulse: a short vertical broadband stripe.

(b) State what the axes represent.

Answer. Horizontal axis: time. Vertical axis: frequency.

(c) State what colour or intensity represents.

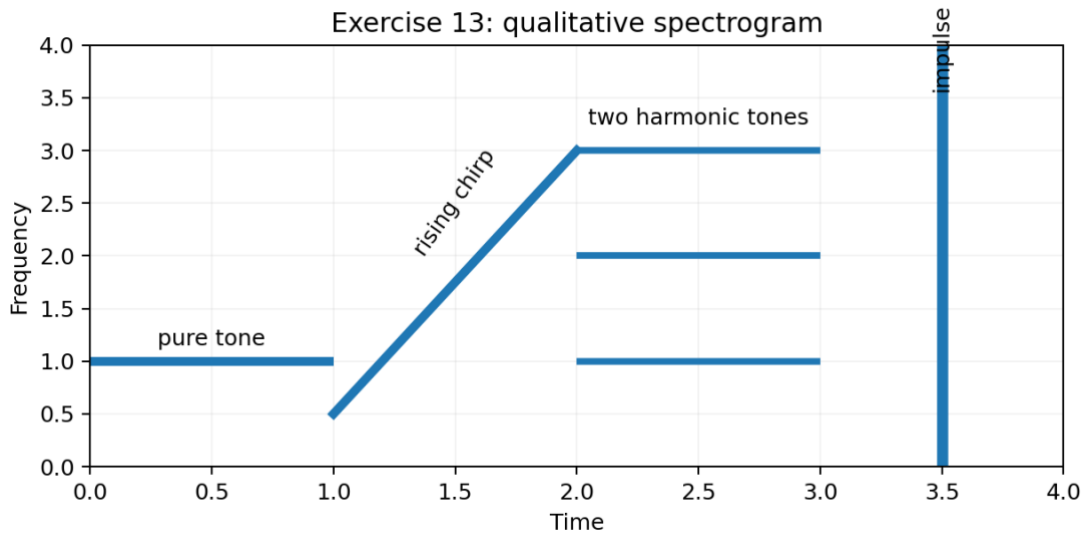
Answer. Spectral magnitude, usually displayed on a logarithmic or decibel scale.

(d) Explain a stationary spectrum versus a time-frequency spectrum.

Answer. A stationary spectrum summarizes frequency content without time location; a spectrogram shows how it changes over time.

(e) Draw a labelled qualitative spectrogram.

Answer. A qualitative drawing is shown below.



Qualitative spectrogram

14. Instantaneous frequency of a chirp

$$x(t) = \cos(2\pi[200t + 300t^2]), \quad 0 \leq t \leq 2 \text{ s.}$$

(a) Identify the phase $\Phi(t)$.

Answer. $\Phi(t) = 2\pi(200t + 300t^2)$.

(b) Compute $\omega_i(t) = d\Phi(t)/dt$.

Answer. $\omega_i(t) = 2\pi(200 + 600t) \text{ rad/s.}$

(c) Compute the instantaneous frequency in hertz.

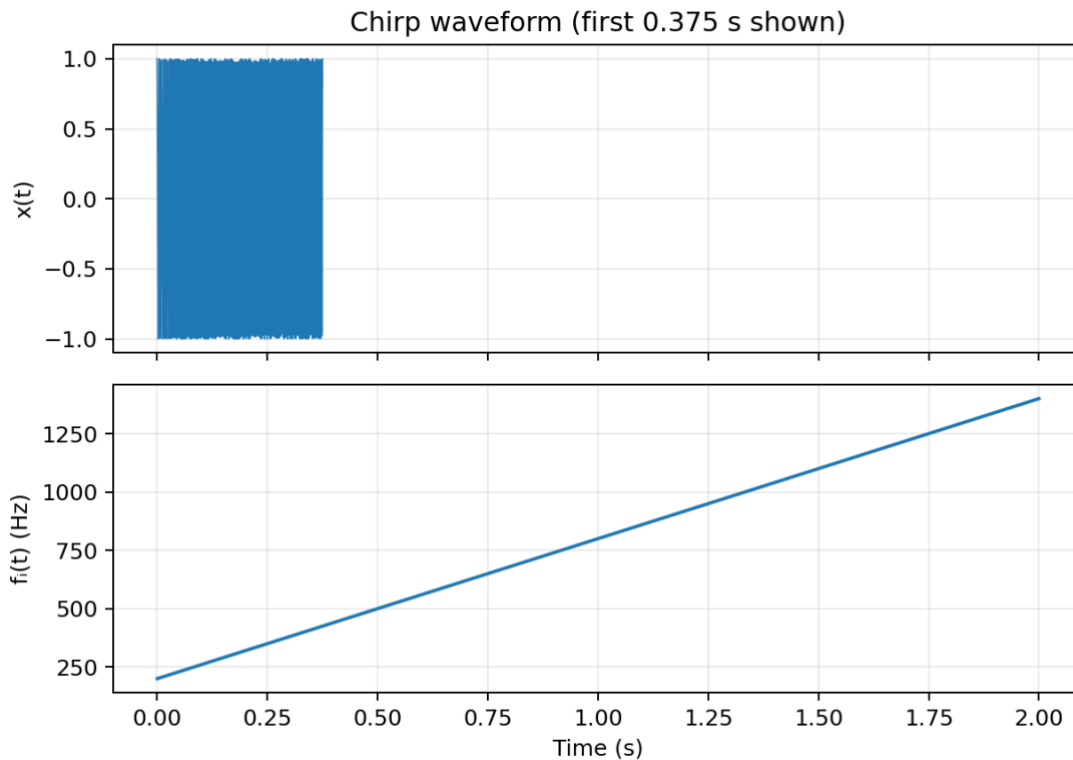
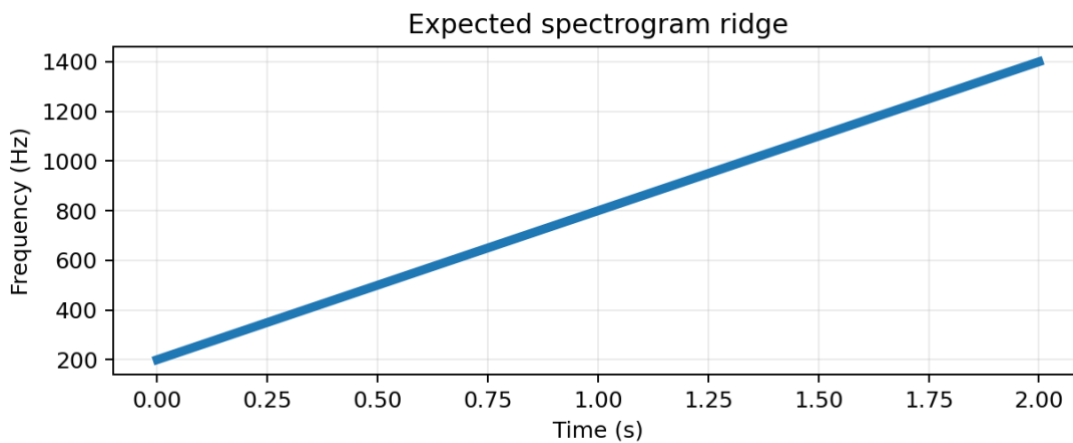
Answer. $f_i(t) = \omega_i(t)/(2\pi) = 200 + 600t \text{ Hz.}$

(d) Determine the starting and ending frequencies.

Answer. $f_i(0) = 200 \text{ Hz}$ and $f_i(2) = 1400 \text{ Hz.}$

(e) Plot $x(t)$ and $f_i(t)$, and sketch the spectrogram ridge.

Answer. The plots and the linear ridge are shown below.

*Chirp waveform and frequency**Chirp ridge*

15. Integrated spectrum-analysis exercise

$$x(t) = 1.5\cos(2\pi 300t) + \cos(2\pi 306t) + [1 + 0.4\cos(2\pi 40t)]\cos(2\pi 900t).$$

(a) Expand the AM term and list every positive-frequency component.

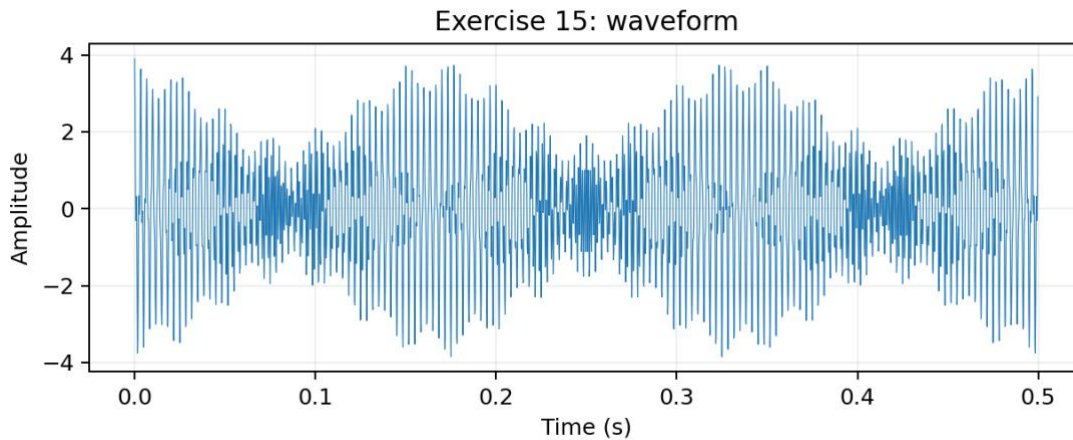
Answer. The AM term is $\cos(2\pi 900t) + 0.2\cos(2\pi 860t) + 0.2\cos(2\pi 940t)$. Positive-frequency cosine components are 300, 306, 860, 900, and 940 Hz.

(b) Construct the complete two-sided spectrum, including magnitudes and phases.

Answer. At ± 300 : magnitude 0.75; at ± 306 : 0.5; at ± 860 and ± 940 : 0.1; at ± 900 : 0.5. All phases are zero.

(c) Plot the waveform over 0.5 s and identify the beat behaviour.

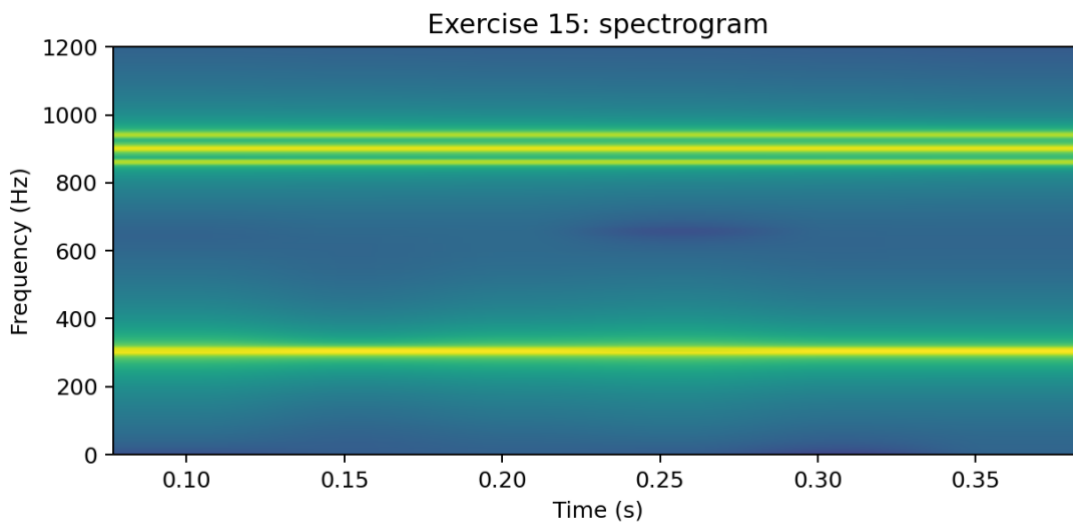
Answer. The plot is shown below. The 300- and 306-Hz components create a 6-Hz beat pattern.



Integrated waveform

(d) Produce a spectrogram and explain the horizontal lines.

Answer. Horizontal components occur near 300, 306, 860, 900, and 940 Hz.



Integrated spectrogram

(e) Replace the 900-Hz carrier by a linear chirp from 700 to 1100 Hz and describe the spectrogram.

Answer. The 900-Hz horizontal carrier becomes a rising diagonal ridge from 700 to 1100 Hz, with two parallel sideband ridges 40 Hz below and above it. The 300- and 306-Hz lines remain horizontal.